

Large-Scale Hypothesis Testing

Progress in statistics is usually at the mercy of our scientific colleagues, whose data is the “nature” from which we work. Agricultural experimentation in the early 20th century led Fisher to the development of analysis of variance. Something similar is happening at the beginning of the 21st century. A new class of “high throughput” biomedical devices, typified by the microarray, routinely produce hypothesis-testing data for thousands of cases at once. This is not at all the situation envisioned in the classical frequentist testing theory of Neyman, Pearson, and Fisher. This chapter begins the discussion of a theory of large-scale simultaneous hypothesis testing now under development in the statistics literature.

2.1 A Microarray Example

Figure 2.1 concerns a microarray example, the *prostate data*. Genetic expression levels for $N = 6033$ genes were obtained for $n = 102$ men, $n_1 = 50$ normal control subjects and $n_2 = 52$ prostate cancer patients. Without going into biological details, the principal goal of the study was to discover a small number of “interesting” genes, that is, genes whose expression levels differ between the prostate and normal subjects. Such genes, once identified, might be further investigated for a causal link to prostate cancer development.

The prostate data is a 6033×102 matrix X having entries¹

$$x_{ij} = \text{expression level for gene } i \text{ on patient } j, \quad (2.1)$$

$i = 1, 2, \dots, N$ and $j = 1, 2, \dots, n$; with $j = 1, 2, \dots, 50$ for the normal controls and $j = 51, 52, \dots, 102$ for the cancer patients. Let $\bar{x}_i(1)$ and $\bar{x}_i(2)$ be the averages of x_{ij} for the normal controls and for the cancer patients.

¹ Obtained from oligonucleotide arrays.

The two-sample t -statistic for testing gene i is

$$t_i = \frac{\bar{x}_i(2) - \bar{x}_i(1)}{s_i}, \quad (2.2)$$

where s_i is an estimate of the standard error of the numerator,

$$s_i^2 = \frac{\sum_1^{50} (x_{ij} - \bar{x}(1))^2 + \sum_{51}^{102} (x_{ij} - \bar{x}(2))^2}{100} \cdot \left(\frac{1}{50} + \frac{1}{52} \right). \quad (2.3)$$

If we had only data from gene i to consider, we could use t_i in the usual way to test the null hypothesis

$$H_{0i} : \text{gene } i \text{ is "null"}, \quad (2.4)$$

i.e., that x_{ij} has the same distribution for the normal and cancer patients, rejecting H_{0i} if t_i looked too big in absolute value. The usual 5% rejection criterion, based on normal theory assumptions, would reject H_{0i} if $|t_i|$ exceeded 1.98, the two-tailed 5% point for a Student- t random variable with 100 degrees of freedom.

It will be convenient for our discussions here to use “ z -values” instead of “ t -values”; that is, we transform t_i to

$$z_i = \Phi^{-1}(F_{100}(t_i)), \quad (2.5)$$

where Φ and F_{100} are the cumulative distribution functions (abbreviated “cdf”) for standard normal and t_{100} distributions. Under the usual assumptions of normal sampling, z_i will have a standard normal distribution if H_{0i} is true,

$$H_{0i} : z_i \sim \mathcal{N}(0, 1) \quad (2.6)$$

(called the *theoretical null* in what follows). The usual two-sided 5% test rejects H_{0i} for $|z_i| > 1.96$, the two-tailed 5% point for a $\mathcal{N}(0, 1)$ distribution.

Exercise 2.1 Plot $z = \Phi^{-1}(F_\nu(t))$ versus t for $-4 \leq t \leq 4$, for degrees of freedom $\nu = 25, 50$, and 100.

But of course we don’t have just a single gene to test, we have $N = 6033$ of them. Figure 2.1 shows a histogram of the N z_i values, comparing it to a standard $\mathcal{N}(0, 1)$ density curve $c \cdot \exp\{-z^2/2\} / \sqrt{2\pi}$, with the multiplier c chosen to make the curve integrate to the same area as the histogram. If H_{0i} were true for every i , that is, if all of the genes were null, the histogram would match the curve. Fortunately for the investigators, it doesn’t: it is too low near the center and too high in the tails. This suggests the presence

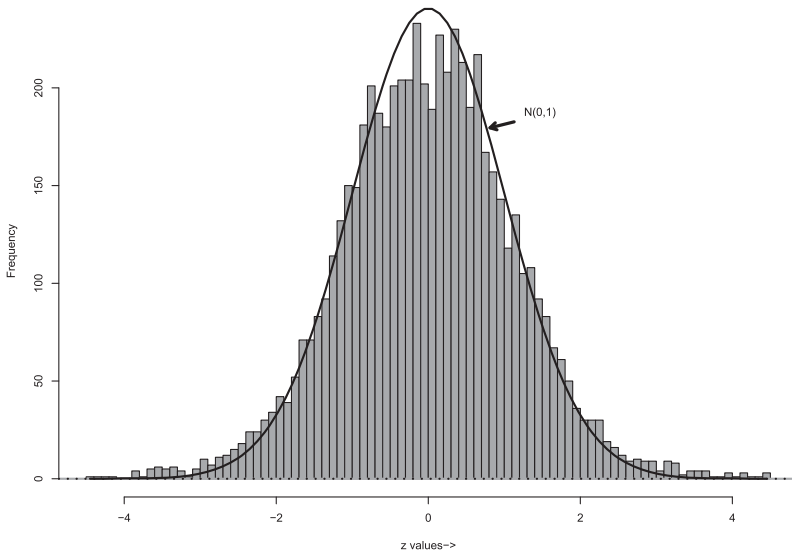


Figure 2.1 Prostate data; z -values testing 6033 genes for possible involvement with prostate cancer; curve is $N(0, 1)$ theoretical null.

of some interesting non-null genes. How to identify the non-null cases dependably, without being misled by the effects of multiple inference, is the subject of intense current research.

A traditional approach to multiple inference uses the Bonferroni bound: we change the rejection level for each test from 0.05 to $0.05/6033$. This amounts to rejecting H_{0i} only if $|z_i|$ exceeds 4.31, rather than 1.96. Now the total probability of mistakenly rejecting even one of the 6033 null hypotheses is less than 5%, but looking at Figure 2.1, 4.31 seems overly cautious. (Only six of the genes are declared non-null.) Empirical Bayes methods offer a less conservative approach to multiple testing.

2.2 Bayesian Approach

The *two-groups model* provides a simple Bayesian framework for multiple testing. We suppose that the N cases (the genes for the prostate study) are each either null or non-null with prior probability π_0 or $\pi_1 = 1 - \pi_0$, and

with z -values having density either $f_0(z)$ or $f_1(z)$,

$$\begin{aligned}\pi_0 &= \Pr\{\text{null}\} & f_0(z) &= \text{density if null} \\ \pi_1 &= \Pr\{\text{non-null}\} & f_1(z) &= \text{density if non-null.}\end{aligned}\quad (2.7)$$

Ordinarily π_0 will be much bigger than π_1 , say

$$\pi_0 \geq 0.90, \quad (2.8)$$

reflecting the usual purpose of large-scale testing: to reduce a vast collection of possibilities to a much smaller set of scientifically interesting prospects. If the assumptions underlying (2.6) are valid, then $f_0(z)$ is the standard normal density,

$$f_0(z) = \varphi(z) = e^{-\frac{1}{2}z^2} / \sqrt{2\pi} \quad (2.9)$$

while $f_1(z)$ might be some alternative density yielding z -values further away from 0.

Let F_0 and F_1 denote the probability distributions corresponding to f_0 and f_1 so that, for any subset $\mathcal{Z}$ of the real line,

$$F_0(\mathcal{Z}) = \int_{\mathcal{Z}} f_0(z) dz \quad \text{and} \quad F_1(\mathcal{Z}) = \int_{\mathcal{Z}} f_1(z) dz. \quad (2.10)$$

The *mixture density*

$$f(z) = \pi_0 f_0(z) + \pi_1 f_1(z) \quad (2.11)$$

has the mixture probability distribution

$$F(\mathcal{Z}) = \pi_0 F_0(\mathcal{Z}) + \pi_1 F_1(\mathcal{Z}). \quad (2.12)$$

(The usual cdf is $F((-\infty, z])$ in this notation, but later we will return to the less clumsy $F(z)$.) Under model (2.7), z has marginal density f and distribution F .

Suppose we observe $z \in \mathcal{Z}$ and wonder if it corresponds to the null or non-null arm of (2.7). A direct application of Bayes rule yields

$$\phi(\mathcal{Z}) \equiv \Pr\{\text{null} | z \in \mathcal{Z}\} = \pi_0 F_0(\mathcal{Z}) / F(\mathcal{Z}) \quad (2.13)$$

as the posterior probability of nullity given $z \in \mathcal{Z}$. Following Benjamini and Hochberg's evocative terminology,² we call $\phi(\mathcal{Z})$ the *Bayes false discovery rate* for $\mathcal{Z}$: if we report $z \in \mathcal{Z}$ as non-null, $\phi(\mathcal{Z})$ is the chance that we've made a false discovery. We will also³ write $\text{Fdr}(\mathcal{Z})$ for $\phi(\mathcal{Z})$.

² Section 4.2 presents Benjamini and Hochberg's original frequentist development of false discovery rates, a name hijacked here for our Bayes/empirical Bayes discussion.

³ A brief glossary of terms relating to false discovery rates appears at the end of the Notes for this chapter.

If $\mathcal{Z}$ is a single point z_0 ,

$$\phi(z_0) \equiv \Pr\{\text{null}|z = z_0\} = \pi_0 f_0(z_0)/f(z_0) \quad (2.14)$$

is the *local Bayes false discovery rate*, also written as $\text{fdr}(z)$.

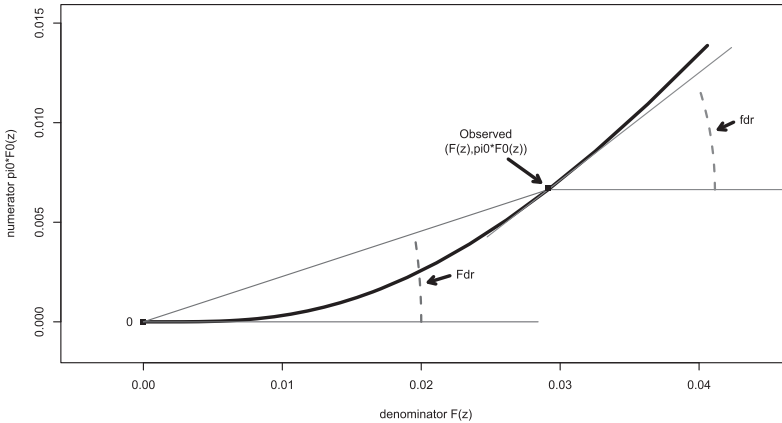


Figure 2.2 Relationship between $\text{Fdr}(z)$ and $\text{fdr}(z)$. $\text{Fdr}(z)$ is the slope of the secant line connecting the origin with the point $(F(z), p_0 \cdot F_0(z))$, the denominator and numerator of (2.16); $\text{fdr}(z)$ is the slope of the tangent line to the curve at that point.

Exercise 2.2 Show that

$$E_f \{\phi(z)|z \in \mathcal{Z}\} = \phi(\mathcal{Z}), \quad (2.15)$$

where E_f indicates conditional expectation with respect to the marginal density $f(z)$. In other words, $\text{Fdr}(\mathcal{Z})$ is the conditional expectation of $\text{fdr}(z)$ given $z \in \mathcal{Z}$.

In applications, $\mathcal{Z}$ is usually a tail interval $(-\infty, z)$ or (z, ∞) . Writing $F(z)$ in place of $F((-\infty, z))$ for the usual cdf,

$$\phi((-\infty, z)) \equiv \text{Fdr}(z) = \pi_0 F_0(z)/F(z). \quad (2.16)$$

Plotting the numerator $\pi_0 F_0(z)$ versus the denominator $F(z)$ shows that $\text{Fdr}(z)$ and $\text{fdr}(z)$ are, respectively, secant and tangent, as illustrated in Figure 2.2, usually implying $\text{fdr}(z) > \text{Fdr}(z)$ when both are small.

Exercise 2.3 Suppose

$$F_1(z) = F_0(z)^\gamma \quad [\gamma < 1] \quad (2.17)$$

(often called *Lehmann alternatives*). Show that

$$\log \left\{ \frac{\text{fdr}(z)}{1 - \text{fdr}(z)} \right\} = \log \left\{ \frac{\text{Fdr}(z)}{1 - \text{Fdr}(z)} \right\} + \log \left(\frac{1}{\gamma} \right), \quad (2.18)$$

and that

$$\text{fdr}(z) \doteq \text{Fdr}(z)/\gamma \quad (2.19)$$

for small values of $\text{Fdr}(z)$.

Exercise 2.4 We would usually expect $f_1(z)$, the non-null density in (2.7), to have heavier tails than $f_0(z)$. Why does this suggest, at least qualitatively, the shape of the curve shown in Figure 2.2?

2.3 Empirical Bayes Estimates

The Bayesian two-groups model (2.7) involves three quantities: the prior null probability π_0 , the null density $f_0(z)$, and the non-null density $f_1(z)$. Of these, f_0 is known, at least if we believe in the theoretical null $\mathcal{N}(0, 1)$ distribution (2.6), and π_0 is “almost known,” usually being close enough to 1 as to have little effect on the false discovery rate (2.13). (In applications, π_0 is often taken equal to 1; Chapter 6 discusses the estimation of both π_0 and $f_0(z)$ in situations where the theoretical null is questionable.) This leaves $f_1(z)$, which is unlikely to be known to the statistician a priori.

There is, however, an obvious empirical Bayes approach to false discovery rate estimation. Let $\bar{F}(\mathcal{Z})$ denote the *empirical distribution* of the N z -values,

$$\bar{F}(\mathcal{Z}) = \# \{z_i \in \mathcal{Z}\} / N, \quad (2.20)$$

i.e., the proportion of the z_i values observed to be in the set $\mathcal{Z}$. Substituting into (2.13) gives an estimated false discovery rate,

$$\overline{\text{Fdr}}(\mathcal{Z}) \equiv \bar{\phi}(\mathcal{Z}) = \pi_0 F_0(\mathcal{Z}) / \bar{F}(\mathcal{Z}). \quad (2.21)$$

When N is large we expect $\bar{F}(\mathcal{Z})$ to be close to $F(\mathcal{Z})$, and $\overline{\text{Fdr}}(\mathcal{Z})$ to be a good approximation for $\text{Fdr}(\mathcal{Z})$.

Just how good is the question considered next. Figure 2.3 illustrates model (2.7): the N values $z_1, z_2, \dots, z_N$ are distributed to the null and non-null arms of the study in proportions π_0 and π_1 . Let $N_0(\mathcal{Z})$ be the number of null z_i falling into $\mathcal{Z}$, and likewise $N_1(\mathcal{Z})$ for the non-null z_i in $\mathcal{Z}$. We can't observe $N_0(\mathcal{Z})$ or $N_1(\mathcal{Z})$, but we do get to see the total number $N_+(\mathcal{Z})$ in $\mathcal{Z}$,

$$N_+(\mathcal{Z}) = N_0(\mathcal{Z}) + N_1(\mathcal{Z}). \quad (2.22)$$

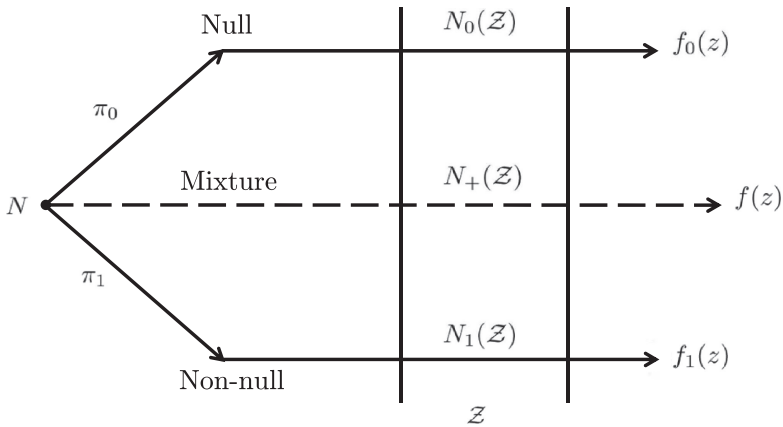


Figure 2.3 Diagram of two-groups model (2.7); N z -values are distributed to the two arms in proportions π_0 and π_1 ; $N_0(\mathcal{Z})$ and $N_1(\mathcal{Z})$ are numbers of nulls and non-nulls in $\mathcal{Z}$; the statistician observes $z_1, z_2, \dots, z_N$ from mixture density $f(z)$ (2.11), and gets to see $N_+(\mathcal{Z}) = N_0(\mathcal{Z}) + N_1(\mathcal{Z})$.

Although $N_0(\mathcal{Z})$ is unobservable, we know its expectation

$$e_0(\mathcal{Z}) \equiv N\pi_0 F_0(\mathcal{Z}). \quad (2.23)$$

In this notation we can express $\overline{\text{Fdr}}(\mathcal{Z})$ as

$$\overline{\text{Fdr}}(\mathcal{Z}) = e_0(\mathcal{Z})/N_+(\mathcal{Z}). \quad (2.24)$$

As an example, the prostate data has $N_+(\mathcal{Z}) = 49$ z_i values in $\mathcal{Z} = (3, \infty)$, that is, exceeding 3; $e_0(\mathcal{Z})$ equals

$$6033 \cdot \pi_0 \cdot (1 - \Phi(3)) \quad (2.25)$$

under the theoretical null (2.6). The upper bound⁴ $\pi_0 = 1$ gives $e_0(\mathcal{Z}) = 8.14$ and

$$\overline{\text{Fdr}}(\mathcal{Z}) = 8.14/49 = 0.166. \quad (2.26)$$

The implication is that about 1/6 of the 49 are false discoveries: if we report the list of 49 to the prostate study scientists as likely prospects, most of their subsequent work will not be in vain. Chapter 4 examines the logic behind this line of thought. Here we will only consider $\overline{\text{Fdr}}(\mathcal{Z})$ as an empirical Bayes point estimator for $\text{Fdr}(\mathcal{Z})$.

⁴ Bounding π_0 by 1 in (2.25) does not imply that $\pi_1 = 0$ in (2.12) since the denominator $\bar{F}(\mathcal{Z})$ in (2.21) is estimated directly from the data regardless of π_0 .

2.4 $\overline{\text{Fdr}}(\mathcal{Z})$ as a Point Estimate

We can express (2.13) as

$$\phi(\mathcal{Z}) = \text{Fdr}(\mathcal{Z}) = e_0(\mathcal{Z})/e_+(\mathcal{Z}), \quad (2.27)$$

where $e_+(\mathcal{Z}) = N \cdot F(\mathcal{Z})$ is the expected total count of z_i values in $\mathcal{Z}$. The quantity we would like to know, but can't observe, is the *false discovery proportion*

$$\text{Fdp}(\mathcal{Z}) = N_0(\mathcal{Z})/N_+(\mathcal{Z}), \quad (2.28)$$

the actual proportion of false discoveries in $\mathcal{Z}$. This gives us three quantities to consider:

$$\overline{\text{Fdr}}(\mathcal{Z}) = \frac{e_0(\mathcal{Z})}{N_+(\mathcal{Z})}, \quad \phi(\mathcal{Z}) = \frac{e_0(\mathcal{Z})}{e_+(\mathcal{Z})}, \quad \text{and} \quad \text{Fdp}(\mathcal{Z}) = \frac{N_0(\mathcal{Z})}{N_+(\mathcal{Z})}. \quad (2.29)$$

The next four lemmas discuss their relationship.

Lemma 2.1 Suppose $e_0(\mathcal{Z})$ as defined in (2.23) is the same as the conditional expectation of $N_0(\mathcal{Z})$ given $N_1(\mathcal{Z})$. Then the conditional expectations of $\overline{\text{Fdr}}(\mathcal{Z})$ and $\text{Fdp}(\mathcal{Z})$ given $N_1(\mathcal{Z})$ satisfy

$$E\{\overline{\text{Fdr}}(\mathcal{Z})|N_1(\mathcal{Z})\} \geq \phi_1(\mathcal{Z}) \geq E\{\text{Fdp}(\mathcal{Z})|N_1(\mathcal{Z})\}, \quad (2.30)$$

where

$$\phi_1(\mathcal{Z}) = \frac{e_0(\mathcal{Z})}{e_0(\mathcal{Z}) + N_1(\mathcal{Z})}. \quad (2.31)$$

Proof Writing $\overline{\text{Fdr}}(\mathcal{Z}) = e_0(\mathcal{Z})/(N_0(\mathcal{Z}) + N_1(\mathcal{Z}))$, Jensen's inequality gives $E\{\overline{\text{Fdr}}(\mathcal{Z})|N_1(\mathcal{Z})\} \geq \phi_1(\mathcal{Z})$. The condition on $e_0(\mathcal{Z})$ is satisfied if the number and distribution of the null case z -values does not depend on $N_1(\mathcal{Z})$. $\square$

Exercise 2.5 Apply Jensen's inequality again to complete the proof.

Note The relationship in (2.30) makes the conventional assumption that $\text{Fdp}(\mathcal{Z}) = 0$ if $N_+(\mathcal{Z}) = 0$.

Lemma 2.1 says that for every value of $N_1(\mathcal{Z})$, the conditional expectation of $\overline{\text{Fdr}}(\mathcal{Z})$ exceeds that of $\text{Fdp}(\mathcal{Z})$, so that in this sense the empirical Bayes false discovery rate is a conservatively biased estimate of the actual

false discovery proportion. Taking expectations over $N_1(\mathcal{Z})$, and reapplying Jensen's inequality, shows that⁵

$$\phi(\mathcal{Z}) \geq E\{\text{Fdp}(\mathcal{Z})\}, \quad (2.32)$$

so that the Bayes Fdr $\phi(\mathcal{Z})$ is an upper bound on the expected Fdp. We also obtain $E\{\overline{\text{Fdr}}(\mathcal{Z})\} \geq E\{\text{Fdp}(\mathcal{Z})\}$, but this is uninformative since $E\{\overline{\text{Fdr}}(\mathcal{Z})\} = \infty$ whenever $\Pr\{N_+(\mathcal{Z}) = 0\}$ is positive. In practice we would use $\overline{\text{Fdr}}^{(\min)}(\mathcal{Z}) = \min(\overline{\text{Fdr}}(\mathcal{Z}), 1)$ to estimate $\text{Fdr}(\mathcal{Z})$, but it is not true in general that $E\{\overline{\text{Fdr}}^{(\min)}(\mathcal{Z})\}$ exceeds $\phi(\mathcal{Z})$ or $E\{\text{Fdp}(\mathcal{Z})\}$.

Exercise 2.6 Show that $E\{\min(\overline{\text{Fdr}}(\mathcal{Z}), 2)\} \geq \phi(\mathcal{Z})$. *Hint:* Draw the tangent line to the curve $(N_+(\mathcal{Z}), \overline{\text{Fdr}}(\mathcal{Z}))$ passing through the point $(e_+(\mathcal{Z}), \phi(\mathcal{Z}))$.

Standard delta-method calculations yield useful approximations for the mean and variance of $\overline{\text{Fdr}}(\mathcal{Z})$, without requiring the condition on $e_0(\mathcal{Z})$ in Lemma 2.1.

Lemma 2.2 Let $\gamma(\mathcal{Z})$ indicate the squared coefficient of variation of $N_+(\mathcal{Z})$,

$$\gamma(\mathcal{Z}) = \text{var}\{N_+(\mathcal{Z})\} / e_+(\mathcal{Z})^2. \quad (2.33)$$

Then $\overline{\text{Fdr}}(\mathcal{Z})/\phi(\mathcal{Z})$ has approximate mean and variance

$$\frac{\overline{\text{Fdr}}(\mathcal{Z})}{\phi(\mathcal{Z})} \sim (1 + \gamma(\mathcal{Z}), \gamma(\mathcal{Z})). \quad (2.34)$$

Proof Suppressing $\mathcal{Z}$ from the notation,

$$\begin{aligned} \overline{\text{Fdr}} &= \frac{e_0}{N_+} = \frac{e_0}{e_+} \frac{1}{1 + (N_+ - e_+)/e_+} \\ &\doteq \phi \left[1 - \frac{N_+ - e_+}{e_+} + \left(\frac{N_+ - e_+}{e_+} \right)^2 \right], \end{aligned} \quad (2.35)$$

and $(N_+ - e_+)/e_+$ has mean and variance $(0, \gamma(\mathcal{Z}))$. $\square$

Lemma 2.2 quantifies the obvious: the accuracy of $\overline{\text{Fdr}}(\mathcal{Z})$ as an estimate of the Bayes false discovery rate $\phi(\mathcal{Z})$ depends on the variability of the

⁵ Benjamini and Hochberg originally used *false discovery rate* as the name of $E\{\text{Fdp}\}$, denoted by FDR; this runs the risk of confusing a rate with its expectation.

denominator $N_+(\mathcal{Z})$ in (2.24).⁶ More specific results can be obtained if we supplement the two-groups model of Figure 2.3 with the assumption of independence,

Independence Assumption:

$$\text{Each } z_i \text{ follows model (2.7) independently.} \quad (2.36)$$

Then $N_+(\mathcal{Z})$ has binomial distribution

$$N_+(\mathcal{Z}) \sim \text{Bi}(N, F(\mathcal{Z})) \quad (2.37)$$

with squared coefficient of variation

$$\gamma(\mathcal{Z}) = \frac{1 - F(\mathcal{Z})}{NF(\mathcal{Z})} = \frac{1 - F(\mathcal{Z})}{e_+(\mathcal{Z})}. \quad (2.38)$$

We will usually be interested in regions $\mathcal{Z}$ where $F(\mathcal{Z})$ is small, giving $\gamma(\mathcal{Z}) \doteq 1/e_+(\mathcal{Z})$ and, from Lemma 2.2,

$$\overline{\text{Fdr}}(\mathcal{Z})/\phi(\mathcal{Z}) \sim (1 + 1/e_+(\mathcal{Z}), 1/e_+(\mathcal{Z})), \quad (2.39)$$

the crucial quantity for the accuracy of $\overline{\text{Fdr}}(\mathcal{Z})$ being $e_+(\mathcal{Z})$, the expected number of the z_i falling into $\mathcal{Z}$. For the prostate data with $\mathcal{Z} = (3, \infty)$ we can estimate $e_+(\mathcal{Z})$ by $N_+(\mathcal{Z}) = 49$, giving $\overline{\text{Fdr}}(\mathcal{Z})/\phi(\mathcal{Z})$ approximate mean 1.02 and standard deviation 0.14. In this case, $\overline{\text{Fdr}}(\mathcal{Z}) = 0.166$ (2.26), is nearly unbiased for the Bayes false discovery rate $\phi(\mathcal{Z})$, and has coefficient of variation about 0.14. A rough 95% confidence interval for $\phi(\mathcal{Z})$ is $0.166 \cdot (1 \pm 2 \cdot 0.14) = (0.12, 0.21)$. All of this depends on the independence assumption (2.36), which we will see in Chapter 8 is only a moderately risky assumption for the prostate data.

Neater versions of our previous results are possible if we add to the independence requirement the relatively innocuous assumption that the number of cases N is itself a Poisson variate, say

$$N \sim \text{Poi}(\eta). \quad (2.40)$$

Lemma 2.3 *Under the Poisson-independence assumptions (2.36) and (2.40),*

$$E\{\text{Fdp}(\mathcal{Z})\} = \phi(\mathcal{Z}) \cdot [1 - \exp(-e_+(\mathcal{Z}))], \quad (2.41)$$

where now $e_+(\mathcal{Z}) = E\{N_+(\mathcal{Z})\} = \eta \cdot F(\mathcal{Z})$.

⁶ Under mild conditions, both $\overline{\text{Fdr}}(\mathcal{Z})$ and $\text{Fdp}(\mathcal{Z})$ converge to $\phi(\mathcal{Z})$ as $N \rightarrow \infty$. However, this is less helpful than it seems, as the correlation considerations of Chapter 7 show. Asymptotics play only a minor role in what follows.

Proof With $N \sim \text{Poi}(\eta)$ in Figure 2.3, well-known properties of the Poisson distribution show that

$$N_0(\mathcal{Z}) \sim \text{Poi}(\eta\pi_0 F_0(\mathcal{Z})) \text{ independently of } N_1(\mathcal{Z}) \sim \text{Poi}(\eta\pi_1 F_1(\mathcal{Z})), \quad (2.42)$$

and

$$N_0(\mathcal{Z})|N_+(\mathcal{Z}) \sim \text{Bi}(N_+(\mathcal{Z}), \pi_0 F_0(\mathcal{Z})/F(\mathcal{Z})) \quad (2.43)$$

if $N_+(\mathcal{Z}) > 0$. But $\pi_0 F_0(\mathcal{Z})/F(\mathcal{Z}) = \phi(\mathcal{Z})$, and

$$\Pr\{N_+(\mathcal{Z}) = 0\} = \exp(-e_+(\mathcal{Z})),$$

giving

$$E\{\text{Fdp}(\mathcal{Z})|N_+(\mathcal{Z})\} = E\left\{\frac{N_0(\mathcal{Z})}{N_+(\mathcal{Z})}|N_+(\mathcal{Z})\right\} = \phi(\mathcal{Z}) \quad (2.44)$$

with probability $1 - \exp(-e_+(\mathcal{Z}))$ and, by definition, $E\{\text{Fdp}(\mathcal{Z})|N_+(\mathcal{Z}) = 0\} = 0$ with probability $\exp(-e_+(\mathcal{Z}))$. $\square$

Applications of large-scale testing often have π_1 , the proportion of non-null cases, very near 0, in which case a region of interest $\mathcal{Z}$ may have $e_+(\mathcal{Z})$ quite small. As (2.39) indicates, $\overline{\text{Fdr}}(\mathcal{Z})$ is then badly biased upward. A simple modification of $\overline{\text{Fdr}}(\mathcal{Z}) = e_0(\mathcal{Z})/N_+(\mathcal{Z})$ cures the bias. Define instead

$$\widetilde{\text{Fdr}}(\mathcal{Z}) = e_0(\mathcal{Z})/(N_+(\mathcal{Z}) + 1). \quad (2.45)$$

Lemma 2.4 Under the Poisson-independence assumptions (2.36) and (2.40),

$$E\{\widetilde{\text{Fdr}}(\mathcal{Z})\} = E\{\text{Fdp}(\mathcal{Z})\} = \phi(\mathcal{Z}) \cdot [1 - \exp(-e_+(\mathcal{Z}))]. \quad (2.46)$$

Exercise 2.7 Verify (2.46).

The Poisson assumption is more of a convenience than a necessity here. Under independence, $\overline{\text{Fdr}}(\mathcal{Z})$ is nearly unbiased for $E\{\text{Fdp}(\mathcal{Z})\}$, and both approach the Bayes false discovery rate $\phi(\mathcal{Z})$ exponentially fast as $e_+(\mathcal{Z})$ increases. In general, $\overline{\text{Fdr}}(\mathcal{Z})$ is a reasonably accurate estimator of $\phi(\mathcal{Z})$ when $e_+(\mathcal{Z})$ is large, say $e_+(\mathcal{Z}) \geq 10$, but can be both badly biased and highly variable for smaller values of $e_+(\mathcal{Z})$.

2.5 Independence versus Correlation

The independence assumption has played an important role in the literature of large-scale testing, particularly for false discovery rate theory, as discussed in Chapter 4. It is a dangerous assumption in practice!

Figure 2.4 illustrates a portion of the *DTI data*, a diffusion tensor imaging study comparing brain activity of six dyslexic children versus six normal controls. (DTI machines measure fluid flows in the brain and can be thought of as generating versions of magnetic resonance images.) Two-sample tests produced z -values at $N = 15\,443$ voxels (three-dimensional brain locations), with each $z_i \sim \mathcal{N}(0, 1)$ under the null hypothesis of no difference between the dyslexic and normal children, as in (2.6).

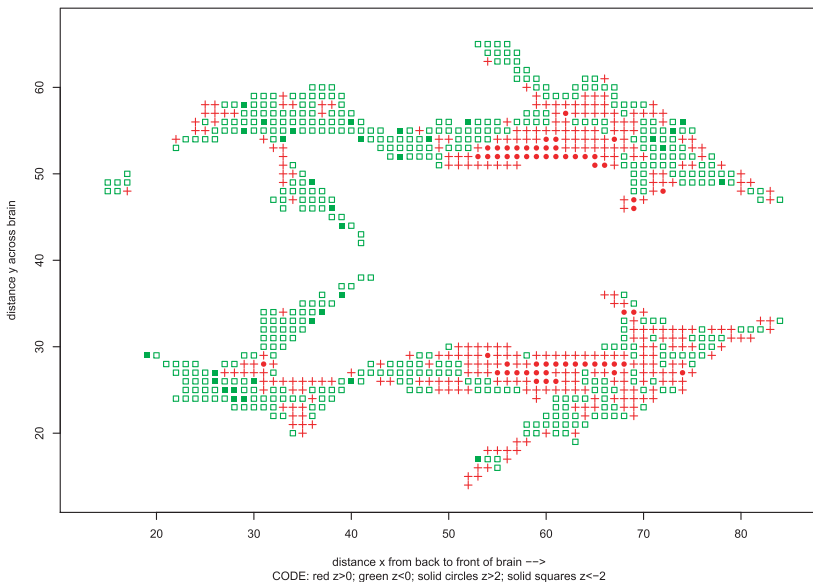


Figure 2.4 Diffusion tensor imaging study comparing brain activity in 6 normal and 6 dyslexic children; z -values for slice of $N = 15\,443$ voxels.

The figure shows only a single horizontal brain section containing 848 of the 15 443 voxels. Red indicates $z_i > 0$, green $z_i < 0$, with solid circles $z_i > 2$ and solid squares $z_i < -2$. Spatial correlation among the z -values is obvious: reds tend to be near other reds, and greens near greens. The symmetric patch of solid red circles near $x = 60$ is particularly striking.

Independence is an obviously bad assumption for the DTI data. We will

see in Chapter 8 that it is an even worse assumption for many microarray studies, though not perhaps for the prostate data. There is no easy equivalent of brain geometry for microarrays. We can't draw evocative pictures like Figure 2.4, but correlation may still be a real problem. Chapter 7 takes up the question of how correlation affects accuracy calculations such as (2.39).

2.6 Learning from the Experience of Others II

Consider the Bayesian *hierarchical model*

$$\mu \sim g(\cdot) \quad \text{and} \quad z|\mu \sim \mathcal{N}(\mu, 1), \quad (2.47)$$

g indicating some prior density for μ , where we will take “density” to include the possibility of g including discrete components. The James–Stein estimator of Figure 1.1 assumes

$$g(\mu) = \mathcal{N}(M, A). \quad (2.48)$$

Model (2.47) also applies to simultaneous hypothesis testing: now take

$$g(\mu) = \pi_0 \Delta_0(\mu) + (1 - \pi_0) g_1(\mu) \quad (2.49)$$

where Δ_0 denotes a delta function at $\mu = 0$, and g_1 is a prior density for the non-null μ_i values. (This leads to a version of the two-groups model having $f_1(z) = \int \varphi(z - \mu) g_1(\mu) d\mu$.) For gene_1 of the prostate data, the “others” in Figure 1.1 are all of the other genes, represented by $z_2, z_3, \dots, z_{6033}$. These must estimate π_0 and g_1 in prior distribution (2.49). Finally, the prior is combined with $z_1 \sim \mathcal{N}(\mu, 1)$ via Bayes theorem to estimate quantities such as the probability that gene_1 is null.

Clever constructions such as $\overline{\text{Fdr}}(\mathcal{Z})$ in (2.21) can finesse the actual estimation of $g(\mu)$, as further discussed in Chapter 11. The main point being made here is that gene_1 is learning from the other genes. “Which others?” is a crucial question, taken up in Chapter 10.

The fact that $g(\mu)$ is much smoother in (2.48) than (2.49) hints at estimation difficulties in the hypothesis testing context. The James–Stein estimator can be quite efficient even for N as small as 10, as in (1.25). Results like (2.39) suggest that we need N in the hundreds or thousands for accurate empirical Bayes hypothesis testing. These kinds of efficiency calculations are pursued in Chapter 7.

Notes

Benjamini and Hochberg's landmark 1995 paper introduced false discovery rates in the context of a now-dominant simultaneous hypothesis testing algorithm that is the main subject of Chapter 4. Efron et al. (2001) recast the *fdr* algorithm in an empirical Bayes framework, introducing the local false discovery rate. Storey (2002, 2003) defined the "positive false discovery rate,"

$$\text{pFdr}(\mathcal{Z}) = E \{N_0(\mathcal{Z})/N_+(\mathcal{Z}) | N_+(\mathcal{Z}) > 0\} \quad (2.50)$$

in the notation of Figure 2.3, and showed that if the z_i were i.i.d. (independent and identically distributed),

$$\text{pFdr}(\mathcal{Z}) = \phi(\mathcal{Z}),$$

(2.27). Various combinations of Bayesian and empirical Bayesian microarray techniques have been proposed, Newton et al. (2004) for example employing more formal Bayes hierarchical modeling. A version of the curve in Figure 2.2 appears in Genovese and Wasserman (2004), where it is used to develop asymptotic properties of the Benjamini–Hochberg procedure. Johnstone and Silverman (2004) consider situations where π_0 , the proportion of null cases, might be much smaller than 1, unlike our applications here.

The two-groups model (2.7) is too basic to have an identifiable author, but it was named and extensively explored in Efron (2008a). It will reappear in several subsequent chapters. The more specialized structural model (2.47) will also reappear, playing a major role in the prediction theory of Chapter 11. It was the starting point for deep studies of multivariate normal mean vector estimation in Brown (1971) and Stein (1981).

The prostate cancer study was carried out by Singh et al. (2002). Figure 2.4, the DTI data, is based on the work of Schwartzman et al. (2005).

The *t*-test (or its cousin, the Wilcoxon test) is a favorite candidate for two-sample comparisons, but other test statistics have been proposed. Tomlins et al. (2005), Tibshirani and Hastie (2007), and Wu (2007) investigate analysis methods that emphasize occasional very large responses, the idea being to identify genes in which a *subset* of the subjects are prone to outlying effects.

The various names for false discovery-related concepts are more or less standard, but can be easily confused. Here is a brief glossary of terms.

Term(s)	Definition
$f_0(z)$ and $f_1(z)$	the null and alternative densities for z -values in the two-groups model (2.7)
$F_0(\mathcal{Z})$ and $F_1(\mathcal{Z})$	the corresponding probability distributions (2.10)
$f(z)$ and $F(\mathcal{Z})$	the mixture density and distribution (2.11)–(2.12)
$\text{Fdr}(\mathcal{Z})$ and $\phi(\mathcal{Z})$	two names for the Bayesian false discovery rate $\Pr\{\text{null} z \in \mathcal{Z}\}$ (2.13)
$\text{fdr}(z)$	the <i>local</i> Bayesian false discovery rate (2.14), also denoted $\phi(z)$
$\text{Fdp}(\mathcal{Z})$	the false discovery proportion (2.28), i.e., the proportion of z_i values in $\mathcal{Z}$ that are from the null distribution
$\bar{F}(\mathcal{Z})$	the empirical probability distribution $\#\{z_i \in \mathcal{Z}\}/N$ (2.20)
$\overline{\text{Fdr}}(\mathcal{Z})$	the empirical Bayes estimate of $\text{Fdr}(\mathcal{Z})$ obtained by substituting $\bar{F}(\mathcal{Z})$ for the unknown $F(\mathcal{Z})$ (2.21)
$\text{FDR}(\mathcal{Z})$	the expected value of the false discovery proportion $\text{Fdp}(\mathcal{Z})$
$N_0(\mathcal{Z})$, $N_1(\mathcal{Z})$, and $N_+(\mathcal{Z})$	the number of null, non-null, and overall z -values in $\mathcal{Z}$, as in Figure 2.3
$e_0(\mathcal{Z})$, $e_1(\mathcal{Z})$, and $e_+(\mathcal{Z})$	their expectations, as in (2.23)